



KALMIX

SCOUT PRO

Product Specification

1. Introduction

1.1 Overview

The KALMIX SCOUT PRO Smart GNSS Receiver is a total solution GNSS receiver, designed based on MediaTek AG3335AT. This positioning application meets strict needs such as car navigation, mapping, surveying, security, agriculture and so on. Only clear view of sky and certain power supply are necessary to the unit. It communicates with other electronic utilities via compatible TYPE-C and saves critical satellite data by built-in backup memory. With low power consumption, the KALMIX SCOUT PRO has 135 channels and DSP accelerators, re-acquires satellite signals in 100 ms and updates position data up to 10Hz.

Trickle-Power allows the unit operates a fraction of the time and Push-to-Fix permits user to have a quick position fix even though the receiver usually stays off.

1.2. Features

G-Mouse provides a host of features that make it easy for integration and use.

1. Use the advanced GNSS module, the module got high performance DSP core (32bit ARM RISC), allow users to design different applications, store in the module, to provide the most economical solution for anybody.
2. High performance tracks up to 135 channels.
3. High sensitivity (-165 dBm) for indoor fixes. The KALMIX SCOUT PRO GNSS module can acquire in only seconds even at low signal levels. As part of KALMIX SCOUT PRO's patented multi-mode GNSS, the KALMIX SCOUT PRO GNSS module can track signal levels as low as -165 dBm. The KALMIX SCOUT PRO supports real-time navigation in urban canyons as well as high sensitivity acquisition needed for indoor environments.
4. Differential capability utilizes real-time RTCM corrections producing 1-

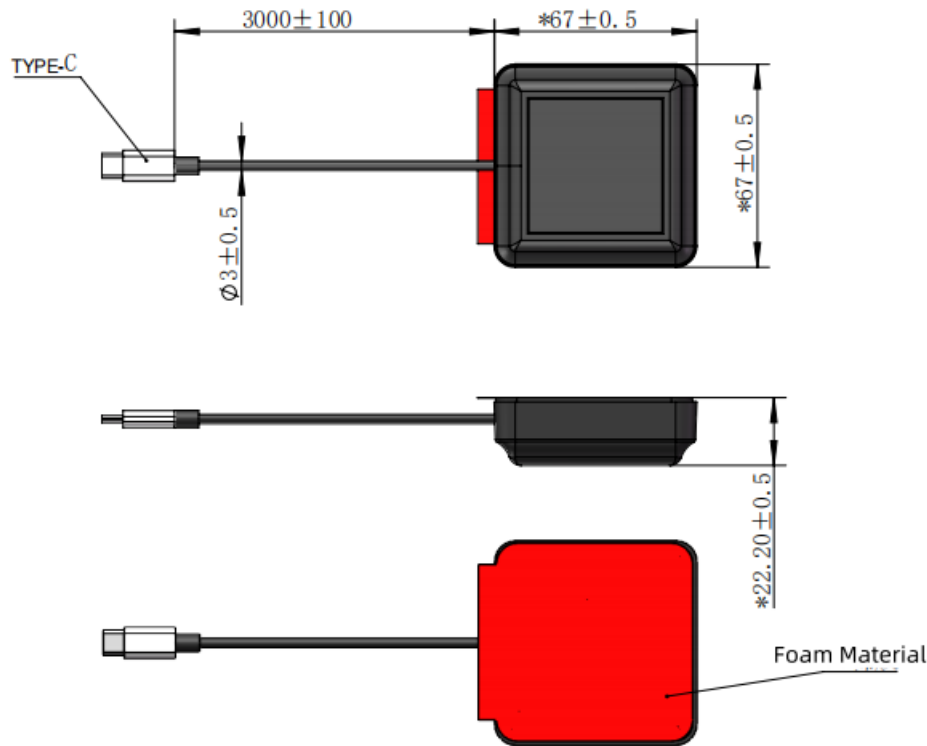
- 5 meter position accuracy (MSM1).
- 5. Compact design ideal for applications with minimal space.
- 6. A rechargeable battery sustains internal clock and memory. The battery is recharged during normal operation.
- 7. Built-in WAAS Demodulator.
- 8. Waterproof IP67 design for industry standard.
- 9. Built-in high gain 33mm*33mm*6mm and 25mm*25mm*4mm ceramic patch-antenna.
- 10. Power on accord with IEC 61000-4-2 (ESD) $\pm 15\text{kV}$ (air), $\pm 8\text{kV}$ (contact)
- 11. Power off accord with IEC 61000-4-2 (ESD) $\pm 25\text{kV}$ (air), $\pm 15\text{kV}$ (contact)
- 12. GNSS reception:
GPS/QZSS: L1 C/A, L5
GLONASS: L1OF
GALILEO: E1(E1B+E1C), E5A
BEIDOU: B1I, B2A
SBAS: WAAS, EGNOS, MSAS, GAGAN

1.3. Technical Specification

1.3.1. Outline, Pin connection

Outline (mm)

KALMIX SCOUT PRO Product Specification



1.3.2. Hardware Interface

The KALMIX SCOUT PRO includes an antenna in a unique style waterproof gadget. Simply connector to one of the accessories linking to your main board or other devices.

PIN	Name	Type	Description	Electrical Characteristics
1	VCC	P	Power in put	DC 5V
2	D-	I/O	TYPE-C Port Data Plus (D-) Signal	
3	D+	I/O	TYPE-C Port Data Plus (D+) Signal	
4	GND	P	GND	

1.3.3. Environmental Characteristics

- 1) Working Temperature: $-40 \sim +85^{\circ}\text{C}$
- 2) Shell and Cable Working Temperature: $-40 \sim +105^{\circ}\text{C}$
- 3) Storage temperature: $-40 \sim +90^{\circ}\text{C}$
- 4) Humidity: $\leq 95\%$

1.3.4. Electrical Characteristics

- 1) Input voltage: 5.0 +/- 10% V DC
- 2) Input current: <80mA
- 3) Backup battery: +3.0V DC (Inner Rechargeable Lithium battery).

1.3.5. Performance

- 1) Channels: 135
- 2) Update rate: 10Hz Maximum
- 3) Acquisition time (average)
Hot start: <1 second(open sky).
Cold start: <35 second(open sky).
- 4) Position accuracy:
Position: <1m CEP
Velocity: 0.01m/sec CEP
Time: 1PPS 25ns
- 5) Dynamic Operation Limit:
Altitude: 18,000m
Velocity: 515m/sec

1.3.6. Interfaces

- 1) Dual channel TYPIC-C compatible level, with user selectable baud rate (Supports baud rate from 9,600 bps to 961,200 bps, 115,200bps by default, 8 bits, no parity bit, 1 stop bit)
- 2) NMEA 0183 Version 4.1 ASCII output (GGA, GSA, GSV, RMC, option GLL, VTG).

2. Operational Characteristics

2.1. Initialization Setup

After the initial self-test is complete, the G-mouse will begin the process of satellite acquisition and tracking. The acquisition process is fully automatic and, under normal circumstances, will take approximately 25

seconds to achieve a position fix (15 seconds if ephemeris data is known). After a position fix has been calculated, valid position and time information will be transmitted over the output channel(s).

The G-Mouse utilizes initial data such as last stored position, data and time as well as satellite orbital data to achieve maximum acquisition performance. If significant inaccuracy exists in the initial data, or if the orbital data is obsolete, it may take a long time to achieve a navigation solution. The G-mouse Auto-locate feature is capable of automatically determining a navigation solution without intervention from the host system. However, acquisition performance can be improved if the host system initializes the G-mouse following the occurrence of one or more of the following events:

- 1) The GNSS receiver is not in use for more than 3 months or transportation over distances further than 500 kilometers.
- 2) Failure of the external memory battery without system standby power.

2.2. Navigation

After the acquisition process is complete, the G-Mouse will begin sending valid navigation information over its output channels. These data include:

- 1) Latitude/longitude/altitude
- 2) Velocity
- 3) Date/time
- 4) Satellite receiver status and Error estimates

Contact US

Shanghai Zhenjie Technology Co., Ltd.

Floor 10, Building 2, No 9, Jinwan Road, Pudong District, Shanghai, China
Service@IOTARS.com